

Observations of interstellar scattering of six pulsars using Polish LOFAR station PL611

Alexandros Filothodoros,^{1,2} Wojciech Lewandowski^{1,2},¹★ Jarosław Kijak,¹★ Bartosz Śmierciak,³ Krzysztof Chyży,³ Leszek Błaskiewicz⁴ and Andrzej Krankowski⁴

¹Janusz Gil Institute of Astronomy, University of Zielona Góra, ul. Szafrana 2, PL-65-516 Zielona Góra, Poland

²The Doctoral School of Exact and Technical Sciences, University of Zielona Góra, al. Wojska Polskiego 69, PL-65-762 Zielona Góra, Poland

³Astronomical Observatory, Jagiellonian University, ul. Orla 171, PL-30-244 Kraków, Poland

⁴Space Radio-Diagnostics Research Centre, University of Warmia and Mazury, ul. Prawocheńskiego 9, PL-10-719 Olsztyn, Poland

Accepted 2024 February 2. Received 2024 February 2; in original form 2023 September 4

ABSTRACT

We present the preliminary results of 4 yr of observations of the scattering of pulsar radiation in the interstellar medium using the Low Frequency Array (LOFAR) PL611 station located in Łazy near Krakow. In this work we show the initial results for six pulsars from our observing campaign. We used the HBA antennas of the station, with a central frequency of 154 MHz and a 72 MHz bandwidth and we were able to detect the frequency-dependent change in the pulse profiles. Splitting the bandwidth into a number of separate sub-bands we obtained independent scatter time estimates, which allowed for the estimation of the scattering frequency scaling slope for each individual observation. Our project has been accumulating data for more than 4 yr and as a result we are in a position to study the time variability of the scattering parameters over this period. We detected significant changes in the scatter time and its frequency scaling for at least two of pulsars. The average values of the frequency scaling index for the six pulsars selected for this work are below the range of values predicted by the simple thin screen model of interstellar scattering. This is in accordance with previous results shown for both higher dispersion measure pulsars observed in the past, as well as the more recent LOFAR observations and other projects similar in the observing frequency. We also discuss the advantages of using individual LOFAR stations (or similar instruments) for this kind of research.

Key words: pulsars: general – ISM: structure.

1 INTRODUCTION

Shortly after the first pulsar discovery (Hewish et al. 1968) it was noticed that such sources are variable and that one reason for this behaviour, is the scattering and scintillation of their signal (Lyne & Rickett 1968; Scheuer 1968) as it propagates through the ionized interstellar medium (ISM). These phenomena arise due to the turbulent nature of the interstellar medium (ISM) which causes variations of the electron density on the path of the traveling radio waves. These density variations cause refraction and diffraction of the propagating wave, and as a result the traveling wavefront is disturbed, which in turn leads to the interference of the wavefront with itself. This may affect the lengths of the propagation path to vary, leading to the scintillation phenomenon (through the existence of multiple images of a pulsar, see Rickett 1990 for a review) and since some of the waves will travel along longer paths the radiation will reach the observer slightly delayed. This causes both temporal and angular broadening of the incoming pulses in a phenomenon known as interstellar scattering (Rickett 1977).

The temporal broadening of the incoming pulse manifests itself in a form of a ‘scattering tail’. This is usually mathematically represented as a convolution of the intrinsic (original) pulse profile with the propagation response function. The exact form of that function depends on the scattering model used (see Williamson 1972, 1973). The most commonly used model is the thin screen model, which assumes that scattering happens in a screen, which is narrow in the direction of propagation and it is placed between the pulsar and the observer (but never close to one or the other). While this model may appear to be a huge simplification of the real phenomenon, one has to take into account that most of the scattering of radio waves will take place in high electron density regions, where the densities may be a few orders of magnitude higher than in the regular interstellar space, and such regions (like H II regions, supernova remnant nebulae, etc.) are indeed much smaller than the pulsar distances, which makes the thin screen approximation viable. This model was shown to agree with observations for many pulsars, especially the objects with relatively large dispersion measures (DMs) – see for example Lewandowski, Kowalińska & Kijak (2015a) and Krishnakumar, Joshi & Manoharan (2017). The validity of the thin screen model was also confirmed by the interstellar scintillation observations, and in particular the detection of scintillation arcs in the secondary spectra (Stinebring et al. 2001; Walker et al. 2004; Cordes et al. 2006). In

* E-mail: boe@astro.ia.uz.zgora.pl (WL); j.kijak@ia.uz.zgora.pl (JK)

the recent years a number of observing programs was devoted to study of these features, utilizing such instruments as Low Frequency Array (LOFAR; Wu et al. 2022), Green Bank and Arecibo telescopes (Stinebring et al. 2022) and the analysis of the data from the Thousand Pulsar Array project performed with MeerKAT (Main et al. 2023). The results of all of these projects have proven that the appearance of the scintillation arcs – which arise due to the existence of thin screen structures in the IISM – is quite prevalent phenomenon in pulsar scintillation observations.

In contrast to the thin screen, the thick screen model assumes irregularities in the ISM occupy a larger part of the line of sight (LOS) and the screen, and this scattering region can be located close to the pulsar or the observer, which results in a different ISM response function. Finally, the extended screen model assumes that the ISM irregularities are spread throughout the whole propagation path, between the pulsar and the observer. Such model assumes that there are no regions of increased electron density along the LOS and scattering appears in the regular ISM, where the ionization fraction may be very low but still non-zero. Such a model may be applicable for nearby pulsars, where a chance of finding an H II region or an ionized nebula along the path is minimal. Currently, there are more complicated models that incorporate multiple scattering screens; however, they are not fully developed yet (Kirsten et al. 2019).

1.1 Scattering frequency dependence

One of the most important ways to study the structure and the turbulent nature of the IISM is the investigation of the frequency dependence of scattering. The amount of observed scattering is highly dependent on the observing frequency, as waves of different wavelength interact differently with different scale turbulence cells in the IISM. The influence of the ISM turbulence on different wavelengths was shown by Rickett (1977), where he argued that there is a power-law relation between the scatter time and the observing frequency, i.e.

$$\tau_d \sim \nu^{-\alpha}, \quad (1)$$

where α is the so-called frequency scaling index or the scattering spectral index. In the model the value of α depends on the spectral index of the spatial electron density distribution β through a relation $\alpha = 2\beta/(\beta - 2)$. If we assume a Kolmogorov spectrum of the density fluctuations then $\beta = 11/3$, which results in $\alpha = 4.4$, and for any homogeneous turbulence model it should not drop below $\alpha = 4.0$ (Romani, Narayan & Blandford 1986). Thus for a thin screen model, the value of α is in the range between 4.0 and 4.4. However, recent studies (Lewandowski, Kowalińska & Kijak 2015a; Lewandowski et al. 2015b, 2018; Geyer & Karastergiou 2016; Kirsten et al. 2019) show that some pulsars exhibit the α values outside this range.

For the majority of the research mentioned above the authors studied not only the observing frequency dependence of scattering for individual pulsars but also the variation of both the amount of scattering (through normalized scatter time estimates) and the slope of the frequency dependence versus the pulsar DM (which is in general a good estimator of pulsar distance). In most of the cases, however, for low DM pulsars the researchers had to resort to using the data from the observations of interstellar scintillation instead of the actual scattering. As these two phenomena are related, there is a tight relation between the scatter time τ and the scintillation's decorrelation bandwidth $\delta\nu_d$:

$$2\pi\tau\delta\nu_d = C_1, \quad (2)$$

where C_1 is a constant of the order of unity, although in general it may depend on the turbulence model or the ISM geometry (Lambert & Rickett 1999) and in some models it is much larger than unity (see for example the finite model in Wu et al. 2023). Using this relation it was possible to calculate the corresponding value of the scatter time from the scintillation measurements. This usually was the only way to estimate the scattering for low-DM pulsars, as for those objects the actual scattering tails only manifest themselves at the lowest frequencies, and in the past most of pulsar observations (and scattering measurements) were performed at frequencies close to or above 300 MHz.

This changed, however, with the development of new radiotelescopes such as LOFAR (van Haarlem et al. 2013). LOFAR pulsar observing capability was already demonstrated by several authors (see for example Stappers et al. 2011; Noutsos et al. 2015; Bilous et al. 2016; Pilia et al. 2016; Bilous et al. 2020), and many of these applications may be extended for use with individual LOFAR stations, although for a considerably smaller sample of brightest pulsars. The core part of LOFAR was also used to study pulsar scattering by Geyer et al. (2017), clearly showing the advantages of using low-frequency observations for the estimation of the scattering properties of nearby pulsars.

The use of an individual LOFAR station for pulsar observations has usually a significant advantage of being able to devote larger amounts of observing time for individual projects due to significantly smaller demand. This also allows us to carry out long-term monitoring projects. The main motivation of our project was the fact that relatively easy access to Polish LOFAR stations allowed us to observe pulsars with the aim to estimate the properties of their scattering – including its frequency dependence – as well as to monitor their variability over extended periods of time. Such projects were extremely rare in the past due to very high observing time demand, and most of the previous studies of pulsar scattering used single scattering measurement at each observing frequency, while with an individual LOFAR station we were able to carry multiple pulsar observations in the so-called ‘local’ mode, i.e. using the stations owner’s time. This allowed us to observe a selected group of pulsars roughly once per month for approximately 5 yr to study the scattering effects in their integrated pulse profiles and will allow us to study the variation of scattering parameters over time. Similar projects were until now performed only for selected individual sources, and more often to study their interstellar scintillation properties and not the variations in their scattering – with one significant exception being the analysis of Long Wavelength Array (LWA) monitoring of scattering for a few pulsars at frequencies below 100 MHz (Bansal et al. 2019). Here we present the preliminary results of our ongoing observational campaign and present the initial results for six sources.

2 OBSERVATIONS, SOURCE SELECTION, AND PRELIMINARY DATA ANALYSIS

The LOFAR is a network of single radio observing stations each capable of observing at low radio frequencies. A single LOFAR station consists of two groups of antennas: the Low Band Antennas (LBAs), which function in the 10–88 MHz range, and the High Band Antennas (HBAs), which operate between 108 and 240 MHz. Both LBA and HBA provide a wide bandwidth of ~ 80 MHz and employ non-moving antennae. LOFAR is a versatile radio telescope van Haarlem et al. (2013). LOFAR stations are spread throughout Europe and can be also used as one interferometric radio telescope or as separate telescopes.

Three single LOFAR stations (PL610, PL611, PL612) are located and functioning in Poland. They belong to the POLFAR Consortium, established in 2007, which consists of a number of Polish educational and research institutions (Błaszkiewicz et al. 2016; Dąbrowski et al. 2016; Błaszkiewicz et al. 2020). Pulsars have been a target for POLFAR stations observations since they became functional in 2015 (Białkowski et al. 2018; Błaszkiewicz et al. 2020).

For this work, we used the PL611 station, located close to Kraków, Poland. This station consists of 96 LBA and 48 HBA tiles (i.e. the station was built in a core-station configuration with only half of the normal station HBA antennas). The pulsar observations are conducted weekly within the so-called stand alone time, which is currently about 30 h a week. This allows for observations of 10 pulsars per week with 3 h integration times, and a total number of 40 separate observations per month, 10 of which (one weekly session every month) are devoted to LBA observations. There are 30 targets observed regularly, for the purposes of our project. In this paper we present the preliminary results for six of these: J0139+5814, J0614+2229, J1543+0929, J1913–0400, J1917+1353, and J2113+4644. For most of these sources the scattering is very prominent at the HBA frequencies, and moreover all of these have relatively simple single-component profiles, which makes scattering analysis easier and less ambiguous.

The pulsar observation pipeline we are using is similar to the one developed by the German LOFAR stations’ Consortium (GLOW)¹ and it was described in Błaszkiewicz et al. (2020). The data acquisition is conducted using the LOFAR und MPIfR pulsare (LUMP) software system.² Stored voltages after analogue station beam-former are coherently dedispersed offline and split into 366 frequency channels (122 channels per data lane) by forming a polyphase filterbank during dedispersion as implemented in DSPSR (van Straten & Bailes 2011). The data are folded using the newest pulsar ephemeris from the ATNF Catalogue³ (Manchester et al. 2005) to form 10-s subintegrations with 1024 bins per period. Further analysis and interference cleaning was made with the help of the PSRCHIVE package (Hotan, van Straten & Manchester 2004), narrow-band interference was removed using a median filter as implemented by the ‘zap’ routine in PSRCHIVE (van Straten, Demorest & Osłowski 2012). In some cases the observations were also manually cleaned of some persistent (mostly wide-band) interference using the interactive subroutine PAZI.

The end-result files of each observation are PSRCHIVE compatible files, with 366 spectral channel data between 118 and 190 MHz, 1024 phase bins for every 10-s subintegration. The data are corrected for DM variations, the value of an actual DM for a given observation was found using the PSRCHIVE subroutine PDMP, and this value was used in further data processing.

2.1 Data preparation and treatment

One of the main goals of our project is the investigation of the frequency dependence of scattering. Pulsar observations conducted using LOFAR can help in that due to the fact that the bandwidth of our observation is relatively large, when compared to the observation’s central frequency: 72 MHz bandwidth with the central frequency of about 150 MHz, which means that the frequency range spans from

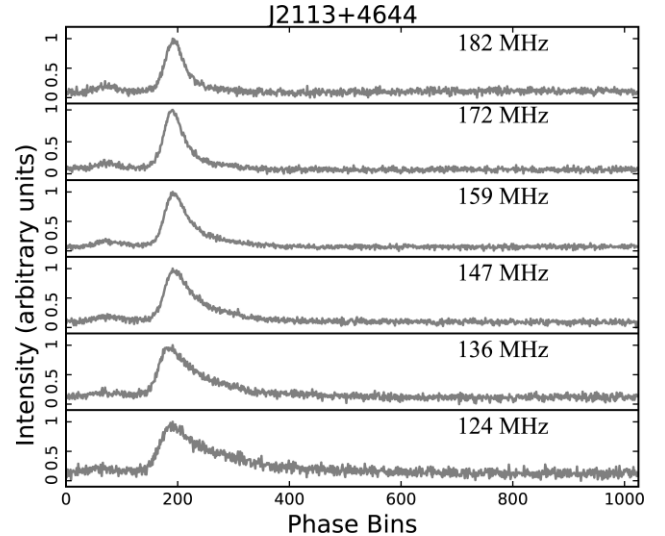


Figure 1. An example of PSR J2113+4644 observation from the PL-611 LOFAR station. The observing bandwidth of 72 MHz was split into six sub-bands. The increase of pulse broadening towards lower frequencies is clearly evident. The frequency-dependent profiles for the remaining five pulsars are shown in Appendix A.

118 to 190 MHz. Since the expected scattering frequency dependence is very steep ($4.0 < \alpha < 4.4$) this in combination with the large fractional bandwidth translates to a huge difference of the scattering between the lowest and the upper edge of the band. The length of the scattering tail should be $1.5^4 \approx 5$ times longer at the low-frequency edge of the bandwidth than at its high-frequency end. This means that one can investigate the scattering frequency dependence using just a single observation and hope to obtain reliable estimate of the frequency scaling index α .

To utilize this we decided to split our bandwidth into separate spectral sub-bands and integrate the pulse profile in each of them to obtain multiple scatter time measurements from each observation, which in turn allows us to check for the frequency scaling. Choosing the number of sub-bands is a compromise between a more reliable estimate of α with more sub-bands, and the decrease of the signal-to-noise ratio (S/N) in every sub-band (which scales as $1/\sqrt{n}$ assuming pulsar flux stays constant thorough the band), which translates into less reliable individual sub-band’s τ measurements. For simplicity we decided to use six frequency sub-bands, since it was easy to implement such split using PSRCHIVE, given 366 spectral channels from three lanes of LOFAR data stream we recorded. At the same time using six sub-bands ensured that even if data in some of them were irrecoverably affected by interference or a low pulsar signal (due to scintillation for example) we were still left with 4–5 (and at minimum 3) usable scatter time estimates. Definitely, for brighter pulsars a larger number of sub-bands is possible, but we left that option for later investigations.

Fig. 1 shows an example set of pulse profiles for six sub-bands of an observation of PSR J2113+4644. The increase of the length of the scattering tail is clearly visible in the lower frequency sub-bands. Similar profile sets for the other five pulsars observed during our project are shown in Appendix A (Figs. A1–A5).

2.2 Modelling of scattering

In the thin screen model the ISM response, or the pulse broadening function (PBF), may be represented as a simple exponential

¹www.mpi-fr-bonn.de/en/lofar/glow

²developed by James Anderson, <https://github.com/AHorneffer/lump-lofar-und-mpi-fr-pulsare/>

³available at <http://www.atnf.csiro.au/research/pulsar/psrcat>.

decay function, with characteristic time-scale τ which is usually called the scatter time. The intrinsic pulse profile is commonly represented as a Gaussian function, and in such case the result of the convolution, i.e. the observed pulse profile, is an exponentially modified Gaussian (EMG) curve. In general the shape of the PBF depends on a scattering model used, and a variety of forms have been proposed in order to describe it. According to the thick screen model $\text{PBF} \sim \sqrt{\pi\tau/(4t^3)} \times \exp(-\pi^2\tau/16t)$ (Williamson 1972). The extended screen model results in $\text{PBF} \sim \sqrt{\pi^5\tau^3/8t^5} \times \exp(-\pi^2\tau/4t)$ (Williamson 1972). If we focus on the thin screen model, $\text{PBF} \sim e^{t/\tau_d}$. Still within the thin screen model framework, some more PBF variations exist, depending on the anisotropy of scattering, for example: $\text{PBF} \sim \tau^{-1}e^{-t/\tau}$ or $\text{PBF} \sim e^{-t/\tau}/\sqrt{\pi\tau t}$ for an isotropic/anisotropic IISM screen respectively (Geyer & Karastergiou 2016).

A few different approaches to modelling of a scattered pulsar profile exist. For example Lewandowski et al. (2013), Lewandowski, Kowalińska & Kijak (2015a), and Lewandowski et al. (2015b) used a simple assumption that the intrinsic pulse profile is a Gaussian function (or a sum of Gaussians corresponding to different profile components) which together with the assumption of a thin screen model translates to the observed profile being an exponentially modified Gaussian, or a sum of such functions. This is then least-squares fitted to the observed profile. According to Löhmer et al. (2001, 2004), Krishnakumar et al. (2015), Krishnakumar, Joshi & Manoharan (2017), and Krishnakumar et al. (2019) a pulse profile observed at higher observing frequency (where the effect of scattering is negligible) can be used in order to create a template. Similarly to the previous method this template has a single Gaussian form or is a sum of Gaussian components and it is convolved with a PBF and possibly other functions that describe the instrumental response and dispersion smearing across spectral channels (if the data are not coherently dedispersed). Such model is then fitted to the observed data using the least-squares algorithm. Further modification of that is the approach used in Geyer & Karastergiou (2016) and Oswald et al. (2021), where the PBF is convolved with a template pulse profile; however, their method offers the extra option of adding an extra term to the model that takes into account the raised profile baseline which may be caused by the very long scattering tails, i.e. in cases of relatively large scattering where the end of the scattering tail contributes to the base level of the following pulse.

While there also exist some completely different approaches to the scatter time measurement, such as the *CLEAN* method deconvolution (Bhat, Cordes & Chatterjee 2003, a method based on the ideas used in interferometric imaging), or the cyclic spectroscopy of the raw voltage data (Demorest 2011), in our project we decided to use the approach from Geyer & Karastergiou (2016) and Oswald et al. (2021), which was already successfully applied to LOFAR data.

To estimate the amount of scattering (by means of the scatter time measurement) we followed the procedure described in Geyer & Karastergiou (2016) and Geyer et al. (2017). This approach, which is a modification of methods used earlier (see for example Lewandowski, Kowalińska & Kijak 2015a; Lewandowski et al. 2015b), assumes that each component of the intrinsic pulsar profile can be approximated by a Gaussian, and that each component's scattering tail is affected by (convolved with) the same PBF function. The result of the convolution is then fitted to the pulse profile using the least-squares method to obtain the best-fitting value of the scatter time τ . For our data we used the simple thin screen model where the scattering tail is just an exponentially decaying function, as implemented in the *GxETRAIN1D* routine in Geyer & Karastergiou (2016).

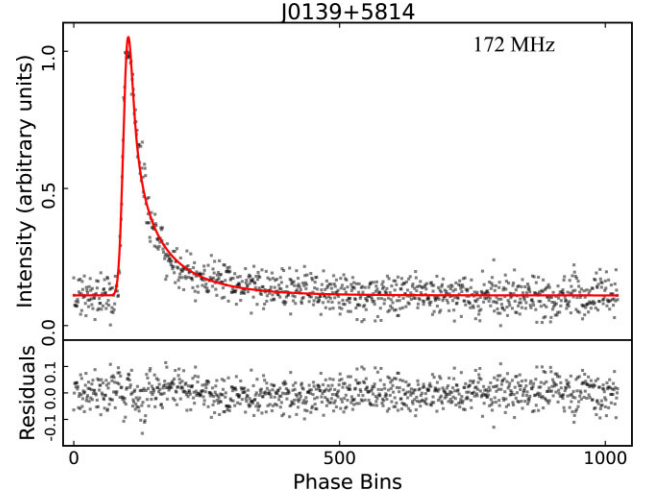


Figure 2. An example fit of a scattering model (the isotropic thin screen model) to one of the sub-band profiles of PSR J0139+5814, using the method described in the text. For the modelling we used the ‘pulse-train’ fitting method from Geyer & Karastergiou (2016) as in this method the baseline of the profile is not removed during the fitting process. This allows to take into account the possibility that an extended scattering tail from the previous pulse may affect the profile baseline.

Additional advantage of using the Geyer & Karastergiou (2016) approach comes for highly scattered pulsars, since this routine takes into account the cases, where the scattering tail is long enough to leak into the next pulse, which may affect the baseline of the pulse. The routine solves the problem by constructing the so-called pulse-train first: the modelling of several consecutive pulses with each next one being affected by the tails from the previous pulses, and the profile shape of one of the later pulses from the train is used to model the observed shape. Fig. 2 shows an example result of the fitting procedure applied to one of the sub-band profiles from an observation of PSR J0139+5814.

After this fitting process, we obtained two scatter time values (one for each isotropic and anisotropic models) for each sub-band of an individual observation, and up to six values total, depending on the interference situation and other factors which deemed some of the profiles too weak or otherwise unusable for fitting. The next step was the estimation of the scatter time frequency scaling index for each observation. For this we followed the approach used in Lewandowski, Kowalińska & Kijak (2015a) and Lewandowski et al. (2015b), which involved fitting a power-law function to the obtained scatter time data:

$$\tau = 10^{-\alpha \log(\nu) + b}, \quad (3)$$

where b is a free parameter that estimates the scattering’s strength at some normalizing frequency (the frequency at which $\log \nu$ is equal to 0, i.e. it is dependent on the units used for the observing frequency), and α is the sought after frequency scaling index. The fit was performed using the Levenberg–Marquardt least-squares method, and following Lewandowski et al. (2015b) we used two methods for the estimation of the α uncertainty: one directly from the L-M fit and the other using the Monte Carlo approach (Goodman & Weare 2010); the larger of the two error estimates was chosen.

We also calculated the scatter time at the standard frequency of 150 MHz (τ_{150}), using equation (3) and the values obtained from an α fit for a given observation. This was done to allow for an investigation of possible variation of both the frequency slope and

the amount of scattering thorough our observing project. We assumed that a value obtained from the interpolation of the α fit will more reliably describe the amount of scattering affecting an observation than any of the individual sub-band τ measurement.

3 RESULTS AND DISCUSSION

In this section, we show the preliminary analysis results of six of our project's targets. As it was mentioned in the previous section we aimed to observe every pulsar in our sample at least once per month, however, due to the observing time limitations that was not always the case. Another factor limiting our ability to estimate the scattering parameters was the radio frequency interference (RFI) situation. In some cases these ruined the observation completely, for example by the presence of prolonged wide-band interference, like coming from thunderstorms or similar. The narrow-band RFI were usually easier to remove from the data, however in some cases, the sheer number of spectral channels that had to be removed was so high that it made some of the sub-bands practically unusable, as the pulse S/N went down enough to prevent a reliable fit for the scatter time. If that happened in three or more sub-bands we rejected the entire observation, as a reliable frequency scaling estimate requires at least three separate sub-band values (from the fitting standpoint) and where possible we were even more strict requiring at least four reliable τ values to qualify the observation for the fit, however that depended on the data quality for a given pulsar.

The RFI obviously affected the weaker pulsars more severely, and it is for that reason, that while for some objects we were able to accumulate over 40 epochs for which we performed the fit for the frequency scaling, for others we were able to gather only about a dozen over the period of almost 4 yr. The last two columns of Table 1 show the number of epochs for which we were able to estimate the frequency scaling and the total time span of our observation, i.e. the time difference between the first and last alpha estimate epochs.

In Table 1 we also show the basic pulsar parameters (pulsar period and DM) from the ATNF Pulsar Catalogue (Manchester et al. 2005) and the weighted average α and τ_{150} values for all six pulsars. The values for individual observations were weighted by the standard method of using the inverse-square of the estimate's uncertainty, and the errors quoted in the table were calculated as the square root of the variance of the weighted average.

Fig. 3 shows an example scatter time versus the observing frequency plot for all the pulsars from our sample. We have chosen a representative single observation for each pulsar, which – in most of the cases – showed the α value close to the weighted average α shown in Table 1. The solid lines on the plots show the result of our fit for the frequency scaling and the dashed lines represent the frequency dependence for the theoretical α value of 4.0 for

reference (anchored to the highest frequency measurement of the scatter time). These plots show the usefulness of the LOFAR wide-band observational capability, as one can clearly see, which – as expected – the difference in the scatter time estimate between the lowest and the highest frequency band is systematic and significant, and the values of τ in the lowest frequency sub-band are usually three to four times higher than at the highest frequency (see Fig. 3).

Fig. 3 and the values quoted in Table 1 clearly show that for five of the six pulsars we have chosen for the purposes of this work the properties of their scattering frequency dependence is significantly different from the theoretical predictions for the frequency scaling, i.e. $4.0 < \alpha < 4.4$, even including the uncertainty estimates.

Figs. 4 and 5 show all of the individual estimates for the scatter time and the frequency scaling index for all the individual observations for the six pulsars presented here, plotted against the observing epoch, which means that they present the variability of the estimated scattering parameters over the course of our project. These estimates allowed us to study the α and τ_{150} values and variations, during a period of almost 4 yr. We performed various statistical tests to check the significance of the observed variation and to differentiate between actual changes and the 'null hypothesis' of the variations representing just a white noise due to the uncertainty of measurements. We performed the Ljung–Box (Ljung & Box 1978), the Augmented Dickey Fuller (Said & Dickey 1984), and the Kwiatkowski–Phillips–Schmidt–Shin tests (Kwiatkowski et al. 1992), and in most cases we were not able to exclude the possibility that the changes observed in the τ_{150} and α values are nothing else than a pure noise. In case of some of the pulsars we observed this may be due to the sparsity of good quality observations which limited the number of usable observational epochs to less than 20 (conf. Table 1), which made some of these tests less suitable for our data. In general a clear variability was detected for PSR J0139+5814 in both the scatter time values and the frequency scaling index. For PSR J1913–0440 we also detected a marginally significant variability in both of these parameters, whereas for PSR J0614+2229 there is no significant variability in the scatter time τ and a hint of variability in α . We discuss our results and compare them to previously published data for each of the individual pulsars below.

3.1 Discussion of the results for individual pulsars

PSR J0139+5814 is a relatively bright pulsar at LOFAR HBA frequencies with the average flux of 197 mJy at 150 MHz, according to the recent LOTASS survey (Sanidas et al. 2019). This pulsar was only sparsely observed for its scattering measurements in the past (see the appendix table in Lewandowski, Kowalińska & Kijak 2015a) and no reliable estimate of the frequency scaling index was found. The average value of the scatter time at 150 MHz we obtained is

Table 1. The basic pulsar parameters and the weighted average scattering parameters for six pulsars from our project. The penultimate column shows the number of individual observational epochs for which we estimated τ_{150} (scatter time interpolated for the frequency of 150 MHz) and its frequency scaling index α . The last two columns show the number of epochs used and the total time span of observations for a given pulsar.

Pulsar	Period (ms)	DM (pc cm ⁻³)	$\langle \tau_{150} \rangle$ (ms)	$\langle \alpha \rangle$	No. of epochs	Time span (yr)
J0139+5814	272.451	73.81	14.6 ± 0.5	3.4 ± 0.2	34	4.81
J0614+2229	334.960	96.91	12.7 ± 0.2	2.2 ± 0.2	27	3.62
J1543+0929	748.448	34.98	9.1 ± 0.3	2.1 ± 0.3	19	2.80
J1913–0440	825.935	89.39	9.1 ± 0.3	3.6 ± 0.2	28	3.85
J1917+1353	194.631	94.54	12.3 ± 0.4	3.3 ± 0.4	16	3.34
J2113+4644	1014.684	141.26	51 ± 1	3.7 ± 0.2	17	3.05

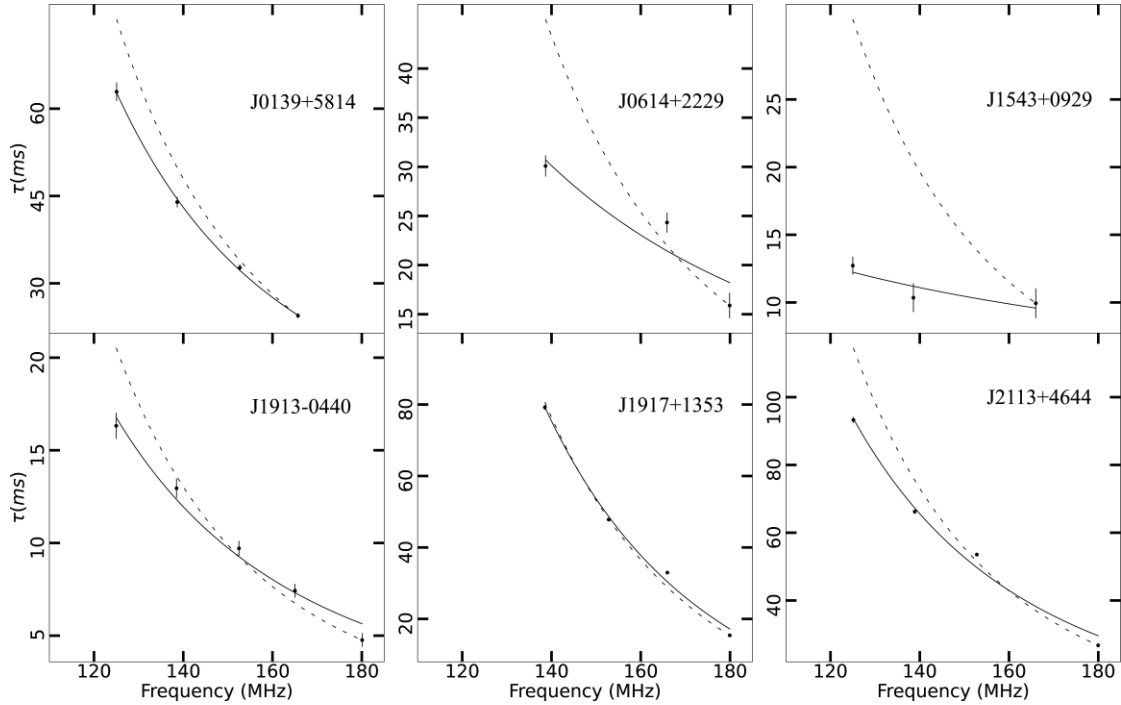


Figure 3. Examples of scatter time versus frequency plots for all of the six pulsars from our sample. Solid line represents the fit for the value of the frequency scaling index α and dashed lines represent the expected dependence with $\alpha = 4.0$. The examples shown are for the observing epochs where the measured value was close to the weighted average value for each pulsar (shown in Table 1).

14.6 ± 0.5 , which is in a relatively good agreement with what can be found in Lewandowski, Kowalińska & Kijak (2015a, see appendix), where they report a measurement of $\tau = 25 \pm 6$ ms at 102 MHz, based upon a pulse profile from the European Pulsar Network (EPN) Database.⁴ The value of the frequency scaling index we obtained is 3.38 ± 0.20 , significantly lower than the theoretical predictions for the thin screen model. Out of the six objects discussed here this is the pulsar with the most striking τ_{150} and α variations (see Figs. 4 and 5). The scatter time measurement changes visibly over the almost 5-yr observing period, between 10 and 20 ms, while the direction of the change switches twice, with a time-scale between 1.5 and 2 yr. For the frequency scaling index one can observe a relatively stable increase over the same period, from a value of $\alpha = 2.5$ to 4.0. No correlation between the τ and α variation was found.

PSR J0614+2229 mean flux at 150 MHz estimated by the LOTASS survey mentioned above is 66 mJy. This pulsar's scattering properties were observed before at similar frequencies. Kuzmin & Losovsky (2007) reported $\tau = 40 \pm 10$ at the frequency of 111 MHz. Later Geyer et al. (2017) analysed the data from LOFAR Core observations of this pulsar and measured the scatter time of 10 ± 1 ms and $\alpha = 1.9 \pm 0.1$ at 150 MHz. Krishnakumar, Joshi & Manoharan (2017) also analysed the scattering for this pulsar. They used the Giant Metrewave Radio Telescope (GMRT) observations, the Ooty Radiotelescope, and the LWA observations with a total frequency range between 148 and 410 MHz and obtained the frequency scaling slope $\alpha = 2.9 \pm 0.1$ and $\tau = 23 \pm 3$ at 148 MHz. Our measurement of the average $\tau_{150} = 12.7 \pm 0.2$ ms is in a relatively good agreement with Geyer et al. (2017), as is the value of $\alpha = 2.2 \pm 0.2$. The values presented by Krishnakumar, Joshi & Manoharan (2017) are,

however, outside of the range of variability of τ and α observed in our data (see Figs. 4 and 5), which is not at all statistically significant for the scatter time and only marginally significant for α (according to some statistical tests we used, see above). The reason for this discrepancy may be the different kind of data used to obtain these values: while we and Geyer et al. (2017) used a single continuous bandwidth observations, Krishnakumar, Joshi & Manoharan (2017) used data from different telescopes at separate observing frequencies, and with much wider frequency range. Their strategy was similar to earlier scattering measurement projects, such as Lewandowski et al. (2013), Lewandowski, Kowalińska & Kijak (2015a), and Lewandowski et al. (2015b), and the possible reasons for erroneous scatter time measurements (which translate later to wrong estimates of α) were widely discussed in these papers. One of such reasons may be the radius-to-frequency mapping (RFM; Kijak & Gil 2003), i.e. the changes to the intrinsic profile width which may simulate scattering-like behaviour, and affect the measured length of the scattering tail, especially at higher frequencies where the scattering is barely noticeable. This effect may cause an underestimate or overestimate the value of the scatter time measurement when the effect is small, compared to the intrinsic profile width, and later it may yield wrong values of α . Therefore we believe that the values obtained in our project (and by Geyer et al. 2017) can be considered to be more reliable than the previous attempts of the study of the scattering for this object, however, there is a strong possibility that the scatter time values measured for PSR J0614+2229 at frequencies above ~ 100 MHz will be heavily affected by the intrinsic profile evolution, and they do not represent the actual scatter time evolution with frequency. In Appendix B (Fig. B1) we show the values of the estimated intrinsic pulse widths for the pulsars from our project, and for PSR J0614+2229 no expected RFM effects can be seen. This indicates that during the modelling of scattering some (or all) of the profile broadening due to the RFM was

⁴<https://www.jb.man.ac.uk/pulsar/Resources/epn/>

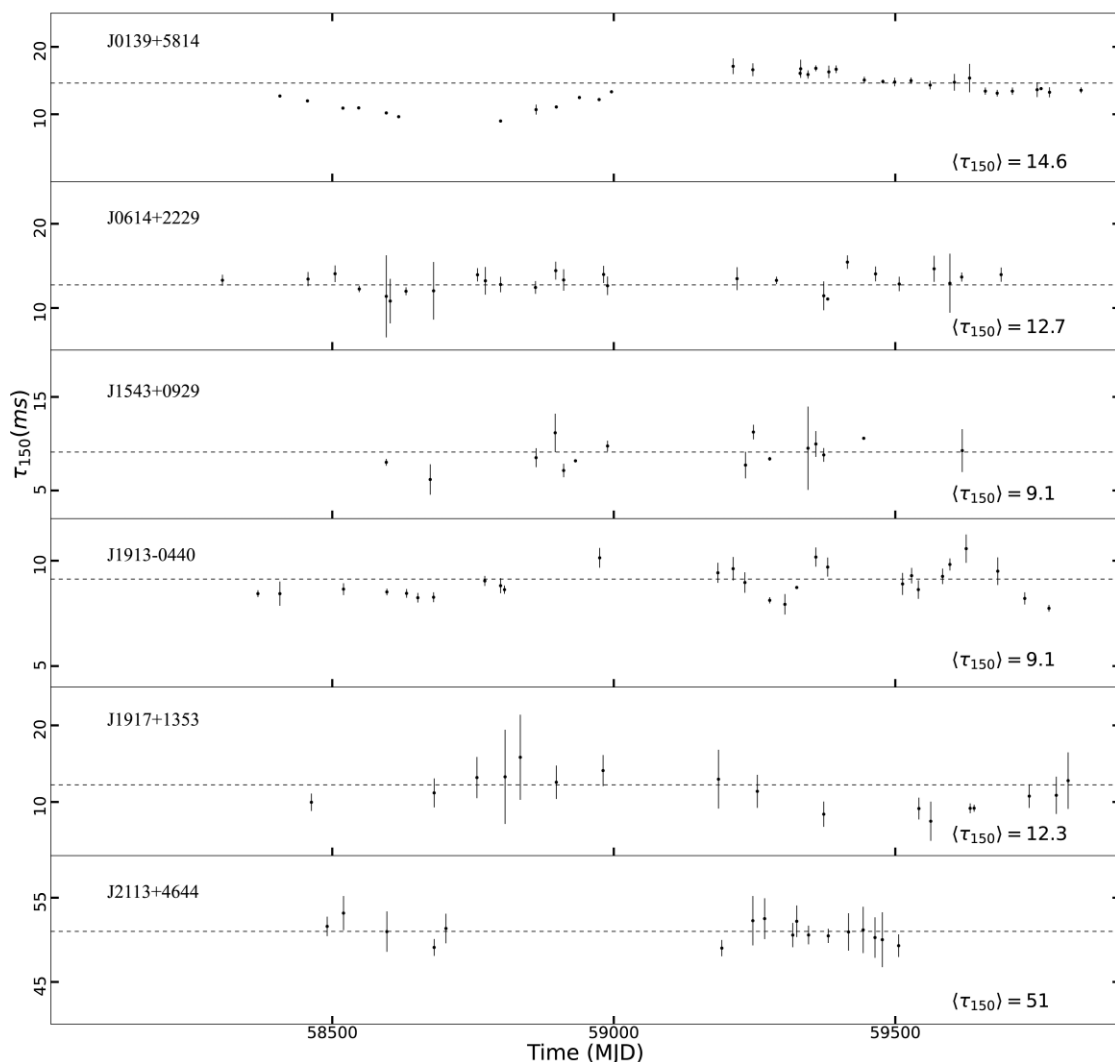


Figure 4. Scatter time estimates for six pulsars from our sample, normalized to the frequency of 150 MHz (τ_{150}) gathered for individual observing epochs over the course of our observing campaign. The dashed line shows the weighted average value from Table 1, which is also quoted in the bottom right of each subplot. A significant variability is clearly noticeable for PSR J0139+5814, PSR J1913–0440, and PSR J1917+1353.

‘absorbed’ into the scattering evolution, yielding erroneous values for scatter time, and consequently the frequency scaling index α . This could be confirmed by lower frequency observations, however, our attempts to detect this pulsar with single station LBA antennas proved unsuccessful.

PSR J1543+0929 is another bright pulsar with a relatively narrow profile, and the mean flux of 570 mJy at 150 MHz (Sanidas et al. 2019). For this pulsar we obtained an unusually low value of $\alpha = 2.1$ and the scatter time τ_{150} of 9.1 ± 0.3 ms. Such a low value of the frequency scaling index suggests that what we measured was not the actual scattering of the pulsar, but rather its profile evolution due to RFM. The values of the intrinsic pulse widths we obtained in our modelling of the scattered profiles for PSR J1543+0929 (see Appendix B) do not show the expected variation due to RFM, which suggests that the pulse width changes were ‘absorbed’ into the scattering parameters during the modelling process. That assumption would be strengthened by the results from Krishnakumar et al. (2019), who reported $\alpha = 3.8 \pm 0.6$ and $\tau_{200} = 0.7 \pm 0.4$, but their estimates are based on very low frequency observations from

LWA only (the frequency range between 52 and 86 MHz). Projecting their values to the LOFAR HBA frequency range one can easily note that the expected scattering at 150 MHz is around 3 ms, much lower than the value we obtained. In that regard it is not surprising that the τ and α variability plots show only a white noise.

PSR J1913–0440, a pulsar with a mean flux of 222 mJy at 150 MHz according to the recent LOTASS survey (Sanidas et al. 2019), was observed frequently in our projects and we managed to gather 29 estimates of the scatter time and frequency scaling index over the period of almost 4 yr (see Table 1). Our average value of τ_{150} is 9.1 ± 0.3 ms and the average frequency scaling index is $\alpha = 3.6 \pm 0.2$. A statistically significant variation of the scatter time can be seen in Fig. 4, especially in the second half of the project, with a quasi-periodic behaviour, which time-scale is below 500 d – however, this does not seem to be correlated (nor anticorrelated) with the pulsar distance from the Sun (which would indicate the possible influence of the interplanetary medium). The variation of the scattering frequency slope α also seems to be statistically significant;

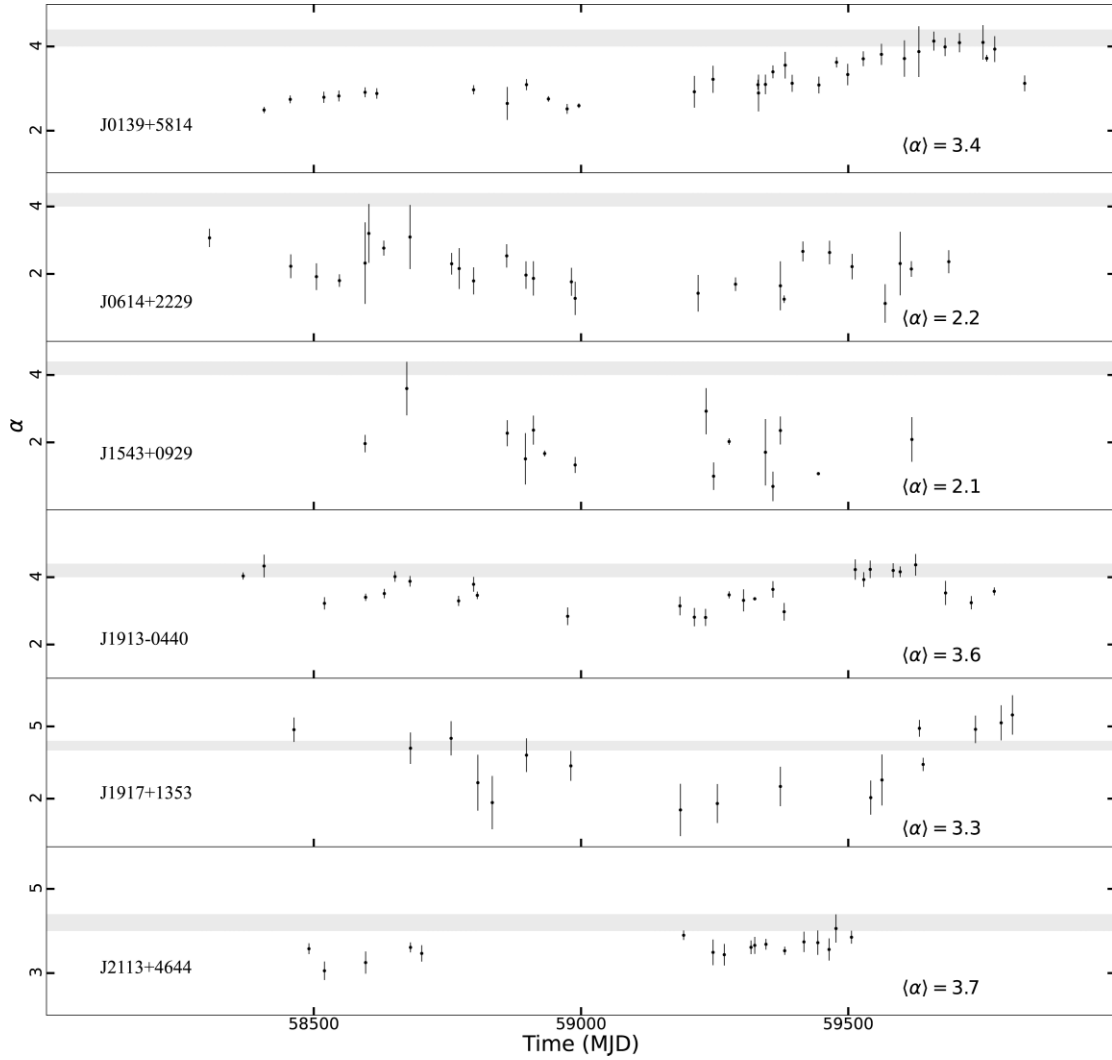


Figure 5. The scattering frequency scaling index α variation over time for six pulsars from our sample. The weighted average value from Table 1 is quoted in the bottom right of each subplot. The grey strip on each plot shows the range of theoretically allowed values for the thin screen isotropic scattering model ($4.0 < \alpha < 4.4$).

however, the changes are much slower than for τ (with a time-scale exceeding 500 d) and they are not correlated with the variations of the scatter time.

The scattering properties of this pulsar were studied before. Kuzmin & Losovsky (2007) reported $\tau_{111} = 35 \pm 15$ ms, and in earlier observations Slee, Dulk & Otrupcek (1980) reported $\tau_{160} = 32 \pm 5$ ms and Alurkar, Bobra & Slee (1986) reported $\tau_{160} = 16.7 \pm 1.8$ ms. The frequency dependence of scattering for this pulsar was also studied in the past: Lewandowski et al. (2015b) combined their own measurements from the GMRT with the data published earlier and estimated $\alpha = 4.18 \pm 0.44$ and $\tau_{1\text{GHz}} = 0.0067$ ms (which given the value of α translates to $\tau_{150} = 18.6$ ms) and a measured value at 150 MHz from GMRT of 19.9 ± 0.8 ms.

This pulsar was also studied by Geyer et al. (2017) who show $\tau_{150} = 7 \pm 0$ and $\alpha = 3.3 \pm 0.1$ based on their LOFAR Core data, and by Krishnakumar et al. (2019) where they presented the values of $\tau_{200} = 4 \pm 3$ ms and $\alpha = 4.5^{+0.3}_{-0.5}$ based on the data from LOFAR, LWA, and ORT. Our data seem to agree with the estimates by Geyer et al. (2017) as the values they reported lay within the range of variations we observed for both α and τ . The values of alpha reported

by Lewandowski et al. (2015b) and Krishnakumar et al. (2019) are significantly larger than what was observed using LOFAR-only data (ours and Geyer et al. 2017), but given the observed variation of α in our project (see Fig. 5) they are still within the range of possible agreement. What is more puzzling is the fact that all the early estimates of the scatter time are much larger than what we observed, with τ_{150} exceeding 30 ms in the early observations (see above the values from Kuzmin & Losovsky 2007 and Slee, Dulk & Otrupcek 1980), and in some other exceeding 15 ms (Alurkar, Bobra & Slee 1986; Lewandowski et al. 2015b). This discrepancy may be due to different methods used for the actual observations and the scatter time estimation; however, one cannot exclude the possibility that over sufficiently long time-scales the scattering properties of this pulsar may change more drastically than what we observed over the 4-yr duration of our project.

PSR J1917+1353, a pulsar with 198 mJy flux at 150 MHz (Sanidas et al. 2019), was studied in the past by various researchers. Slee, Dulk & Otrupcek (1980) published the scatter time value $\tau_{160} = 12 \pm 3$ ms, later Alurkar, Bobra & Slee (1986) reported $\tau_{160} = 11.7 \pm 1.9$ ms and Kuzmin & Losovsky (2007) reported a value of

$\tau_{111} = 40 \pm 20$ ms. Based on these estimates and their own analysis of EPN data base of pulse profiles Lewandowski, Kowalińska & Kijak (2015a) estimated the frequency scaling of this pulsar to be $\alpha = 2.11 \pm 0.05$; however, they deemed this value to be unreliable. Geyer et al. (2017) studied the LOFAR Core observations data for this pulsar and found the scatter time at 150 MHz of 11 ± 1 ms and $\alpha = 2.8 \pm 0.4$. These values are in a good agreement with our results (see Table 1); however, our measurement of the frequency slope at $\alpha = 3.3 \pm 0.4$ is slightly closer to the theoretically predicted range. The variability of the scatter time (see Fig. 4) seems to indicate changes happening on a time-scale exceeding 500 d; however, due to relatively large τ_{150} uncertainty estimates for some of the observational epochs and a relatively low number of 16 observations all the statistical methods we applied struggle with the recognition of this variability as genuine changes. Similarly the variability of α we show in Fig. 5 may be questionable due to its enormous change in values between 2.5 and going up all the way to over 5.0.

PSR J2113+4644 is the highest DM pulsar in our sample (141 pc cm^{-3}) and both the scattering itself and its change with the observing frequency are clearly evident (conf. Fig. 1). This pulsar is relatively bright (244 mJy at 150 MHz, Sanidas et al. 2019), and we were able to observe it 17 times over the period of slightly over 3 yr. No significant variability was detected in the scattering parameters, with the average values of $\tau_{150} = 51 \pm 1$ ms and $\alpha = 3.7 \pm 0.2$. This pulsar was only sparsely observed before at similar frequencies: Kuzmin & Losovsky (2007) reported $\tau_{111} = 120 \pm 40$, and Krishnakumar, Joshi & Manoharan (2017) reported $\tau_{148} = 50 \pm 2$ ms, and the frequency scaling index of 4.2 ± 0.1 based on observations with a total frequency range of 148–327 MHz.

4 SUMMARY AND CONCLUSIONS

We presented and discussed the initial results of the interstellar scattering measurements for six pulsars from our ongoing observing campaign conducted with the PL611 LOFAR station. For all of the six pulsars for which the average parameters are presented in Table 1 the frequency scaling index comes significantly below the range allowed by the simple scattering theory ($4.0 < \alpha < 4.4$). Such discrepancy has been observed before for higher DM and more distant pulsars (see for example Lewandowski, Kowalińska & Kijak 2015a; Krishnakumar, Joshi & Manoharan 2017) as well as in lower frequency observations (Geyer et al. 2017; Bansal et al. 2019; Krishnakumar et al. 2019). In Lewandowski et al. (2013) the authors discussed a variety of possible causes of such deviation, which included the non-Kolmogorov character of the IISM plasma turbulence, anisotropy of scattering, and possibly the instances of multiple scattering events along the LOS. The latter suggestion is still being strongly considered, especially after the observations of multiple scintillation arcs in several pulsars, which would suggest multiple scatterings along the LOS (see for example Main et al. 2023). Such a possibility may be well justified, especially for the high DM (distant) pulsars that were showing the α deviation based on higher frequency observation, as the chance for the multiple scattering to occur increases with the pulsar distance. Surprisingly, however, some of the multiple-arc pulsars presented by Main et al. (2023) are actually relatively nearby objects. This would suggest that a chance for multiple scatterings even for low-DM objects is not negligible. A significant difficulty in considering multiple scatterings as a possible explanation of the observed deviation of the scaling index α is the fact that the models of such scattering are not fully developed and our understanding of the influence of multiple scatterings on the observed pulse profile shapes is very limited.

The remaining two possibilities should also be considered: the non-Kolmogorov nature of the IISM turbulence and the anisotropy and/or inhomogeneity of scattering, with the latter one being more likely in our opinion. The possible effects of inhomogeneity of the IISM were considered as possible sources of the deviation of α by Cordes & Lazio (2001) and Wu et al. (2023). As it was shown by Geyer & Karastergiou (2016), the anisotropy of scattering can cause changes to the shape of the observed scattering tails, which can translate to erroneous scatter time estimates and that in turn will affect the frequency scaling index estimates.

Additionally, these anisotropy/inhomogeneity effects may – in certain circumstances – cause the variation in the observed profile shapes. This may be due to the motion of the LOS caused by the motion of the pulsar, the Solar System and/or the IISM region itself (and the anisotropy present in it). In such situation the effects of the anisotropy on the observed profile, and by extension the scatter time measured using standard model as well as the resulting scattering frequency dependence slope, can change over time. This can be only detected in long-term monitoring projects such as our observing campaign.

Furthermore, the observed variations in the scattering parameters may be happening due to the genuine changes of the properties of scattering along the LOS, other than a sharp anisotropy/truncated screens mentioned above. The IISM regions that the LOS crosses may be characterized by different electron density (i.e. it may be anisotropic) but may also exhibit slightly different turbulence properties (i.e. inhomogeneity), which would translate into a variable value of the observed scattering frequency scaling index. Monitoring of such variations would give us an insight into the spatial scales of the electron density and turbulence profile fluctuations, since in most cases we will be able to translate the time-scale of the observed variability into the linear scale through the pulsar proper motion. The knowledge of the scattering screen location may be useful here, but that is where the observations of scintillation arcs may be invaluable (Cordes et al. 2006).

This information may be used along the observations of DM variations, which are our most studied source of information about the change in the electron density along the LOS (see for example Hobbs et al. 2004; Donner et al. 2020). While the value of DM provides information about the total electron column density between the pulsar and the observer, using a single scattering screen model we can assume that most of the observed DM variability should be attributed to the screen itself. The pulsar LOS moving across a screen with a non-uniform electron density (or a non-uniform depth) will cause the observed DM to vary. On the other hand the strength of the scattering phenomenon does not depend directly on the value of the electron density (n_e), but on its small-scale fluctuation (δn_e). This causes the scatter times measured for different pulsars to only loosely correlate with DM, as the higher value of n_e may (but does not have to) translate to larger density fluctuations. This is reflected in the scatter time – DM relation for the pulsar population (see for example Lewandowski, Kowalińska & Kijak 2015a) which shows a relatively large scatter.

Therefore the study of the magnitude of the scattering parameters variability may provide additional information regarding the structure of the ISM that would supplement the results obtained from DM variation studies. For example, if the positive correlation between DM changes and the scatter time variability is found, that may indicate that the DM variation may be caused by the increase of the electron density along the LOS in the screen; on the other hand a lack of such correlation may point to more of a geometrical cause, i.e. if the LOS crosses a slightly wider part of the screen the

value of DM may increase, while the scattering properties would remain mostly unchanged. A more detailed study of the effects of all the factors that may influence the observed DM/scattering variability would obviously need a more strict approach in an updated scattering and inhomogeneity model, but this would be well beyond the scope of this paper.

The kind of studies mentioned above would require a high degree of certainty on the values of the measured scatter times, which means ensuring that the scattering model used for the pulse broadening fits to find scatter time is correct. For that reason we will analyse our data using other scattering models (anisotropic thin screen, thick screen, etc.) and we will present our results in a forthcoming paper.

For the six pulsars presented in this work we have gathered a significant number of measurements of the pulse broadening effects and the estimations of the frequency dependence of scattering. This allows us to study the variability of these parameters over the course of the four year duration of our project. We detected two clear cases of variability, for pulsars J0139+5814 and J1913–0440, and a hint of possible genuine variation in two other objects (PSR J1917+1353 and PSR J0614+2229; the latter only shows some variation in the frequency scaling index). A more detailed study of the observed variations of the DM and the scattering parameters for the six pulsars presented here, as well as for the other sources observed during our campaign (including the sources observed in the LBA frequency range) is currently in preparation.

Our results clearly confirm that an individual LOFAR station is capable of this kind of research. As for the average values of the scatter time τ and the frequency scaling index α we obtained our results are in agreement with the earlier observations of these six pulsars, or to be more precise – with the more modern ones, which were performed with the current instruments, receivers, analysis methods and at similar observing frequencies.

There are two main advantages of using LOFAR and its individual telescopes (stations) for this kind of research. The first one is the ability to measure the properties of the scattering frequency scaling from a single observation by means of bandwidth splitting. This method is used widely for scattering measurements with LOFAR and LWA (Geyer et al. 2017; Bansal et al. 2019; Krishnakumar et al. 2019) and our results confirm that individual LOFAR stations are also able to contribute in that regard, at least for the brighter pulsars. In the research of the frequency scaling properties of interstellar scattering such approach, where applicable, is clearly superior to combining individual pulse broadening measurements performed at several isolated frequencies, with different telescopes (and different hardware) and at different times. The later is especially important when one considers the possibility that the scattering parameters may change with time (as we present here for at least two objects). Such time variability in case of the scattering projects performed in the past may have significantly affected the estimates of the scatter time, and more importantly for the frequency scaling index α .

The second advantage is specific to the use of an individual LOFAR station, for which there is usually less demand for the observing time, which allows to conduct long-term monitoring projects of astrophysical objects, such as the observing campaign we are carrying. This monitoring approach allows us to use new avenues of research, which are usually not possible with larger telescopes, where observing time demand is much higher. The study of the ISM inhomogeneity and anisotropy effects which can manifest themselves as time variation of the scattering parameters (see Briskin et al. 2010) is definitely one example of such studies. Research projects of this kind were conducted only very rarely in the past, usually only for an individual target, and most of the time they were

using scintillation studies rather than scattering measurements. One significant exception in this regard is the work of Bansal et al. (2019), who observed seven pulsars over 3 yr at frequencies below 100 MHz.

As we mentioned above more detailed analysis of the observed variability and the results for other pulsars included in our project, as well as the application of different scattering models will be described in a forthcoming publication.

ACKNOWLEDGEMENTS

This work was supported by grant 2018/29/B/ST9/02569 of the National Science Centre, Poland.

AF is a PhD candidate of the Doctoral School of Exact and Technical Sciences, University of Zielona Góra.

We would like to thank the Ministry of Education and Science of Poland for granting funds for the Polish contribution to the International LOFAR Telescope, LOFAR2.0 upgrade (decision number: 2021/WK/2) and for maintenance of the LOFAR PL-610 Borówiec, LOFAR PL-611 Łazy, LOFAR PL-612 Bałdy stations (decision numbers: 30/530252/SPUB/SP/2022, 29/530358/SPUB/SP/2022, 28/530020/SPUB/SP/2022, respectively).

We would like to thank the Computer Centre of the University of Zielona Góra for the assistance with the setup of the pulsar data acquisition and analysis system and for the ongoing assistance with its maintenance.

DATA AVAILABILITY

The data underlying this article will be shared on reasonable request to the corresponding author.

REFERENCES

- Alurkar S. K., Bobra A. D., Slee O. B., 1986, *Aust. J. Phys.*, 39, 433
- Bansal K., Taylor G. B., Stovall K., Dowell J., 2019, *ApJ*, 875, 146
- Bhat N. D. R., Cordes J. M., Chatterjee S., 2003, *ApJ*, 584, 782
- Białkowski S., Lewandowski W., Kijak J., Błaszkiwicz L., Krankowski A., Osłowski S., 2018, *Ap&SS*, 363, 110
- Bilous A. V. et al., 2016, *A&A*, 591, A134
- Bilous A. V. et al., 2020, *A&A*, 635, A75
- Błaszkiwicz L., Lewandowski W., Krankowski A., Kijak J., Koralewska O., Dąbrowski B. P., 2016, *Acta Geophys.*, 64, 293
- Błaszkiwicz L. et al., 2020, in Małek K., Polińska M., Majczyna A., Stachowski G., Poleski R., Wyrzykowski Ł., óżańska A. eds, Vol. 10, XXXIX Polish Astronomical Society Meeting, Polish Astronomical Society. p. 46
- Briskin W. F., Macquart J. P., Gao J. J., Rickett B. J., Coles W. A., Deller A. T., Tingay S. J., West C. J., 2010, *ApJ*, 708, 232
- Cordes J. M., Lazio T. J. W., 2001, *ApJ*, 549, 997
- Cordes J. M., Rickett B. J., Stinebring D. R., Coles W. A., 2006, *ApJ*, 637, 346
- Demorest P. B., 2011, *MNRAS*, 416, 2821
- Dąbrowski B. P., Krankowski A., Błaszkiwicz L., Rothkaehl H., 2016, *Acta Geophys.*, 64, 825
- Donner J. Y. et al., 2020, *A&A*, 644, A153
- Geyer M., Karastergiou A., 2016, *MNRAS*, 462, 2587
- Geyer M. et al., 2017, *MNRAS*, 470, 2659
- Goodman J., Weare J., 2010, *Commun. Appl. Math. Comput. Sci.*, 5, 65
- Hewish A., Bell S. J., Pilkington J. D. H., Scott P. F., Collins R. A., 1968, *Nature*, 217, 709
- Hobbs G., Lyne A. G., Kramer M., Martin C. E., Jordan C., 2004, *MNRAS*, 353, 1311
- Hotan A. W., van Straten W., Manchester R. N., 2004, *PASA*, 21, 302
- Kijak J., Gil J., 2003, *A&A*, 397, 969

- Kirsten F., Bhat N. D. R., Meyers B. W., Macquart J. P., Tremblay S. E., Ord S. M., 2019, *ApJ*, 874, 179
- Krishnakumar M. A., Mitra D., Naidu A., Joshi B. C., Manoharan P. K., 2015, *ApJ*, 804, 23
- Krishnakumar M. A., Joshi B. C., Manoharan P. K., 2017, *ApJ*, 846, 104
- Krishnakumar M. A., Maan Y., Joshi B. C., Manoharan P. K., 2019, *ApJ*, 878, 130
- Kuzmin A. D., Losovsky B. Y., 2007, *Astron. Astrophys. Trans.*, 26, 597
- Kwiatkowski D., Phillips P. C., Schmidt P., Shin Y., 1992, *J. Econometr.*, 54, 159
- Lambert H. C., Rickett B. J., 1999, *ApJ*, 517, 299
- Lewandowski W., Dembska M., Kijak J., Kowalińska M., 2013, *MNRAS*, 434, 69
- Lewandowski W., Kowalińska M., Kijak J., 2015a, *MNRAS*, 449, 1570
- Lewandowski W., Rożko K., Kijak J., -tacharyya B., Roy J., 2015b, *MNRAS*, 454, 2517
- Lewandowski W., Kijak J., Błaszczewicz L., Rożko K., Krankowski A., 2018, in Weltevrede P., Perera B. B. P., Preston L. L., Sanidas S.eds, Vol. 337, Pulsar Astrophysics the Next Fifty Years, Cambridge University Press, United Kingdom, p. 356
- Ljung G., Box G., 1978, *Biometrika*, 65
- Löhmer O., Kramer M., Mitra D., Lorimer D. R., Lyne A. G., 2001, *ApJ*, 562, L157
- Löhmer O., Mitra D., Gupta Y., Kramer M., Ahuja A., 2004, *A&A*, 425, 569
- Lyne A. G., Rickett B. J., 1968, *Nature*, 218, 326
- Main R. A. et al., 2023, *MNRAS*, 518, 1086
- Manchester R. N., Hobbs G. B., Teoh A., Hobbs M., 2005, *AJ*, 129, 1993
- Noutsos A. et al., 2015, *A&A*, 576, A62
- Oswald L. S. et al., 2021, *MNRAS*, 504, 1115
- Pilia M. et al., 2016, *A&A*, 586, A92
- Rickett B. J., 1977, *ARA&A*, 15, 479
- Rickett B. J., 1990, *ARA&A*, 28, 561
- Romani R. W., Narayan R., Blandford R., 1986, *MNRAS*, 220, 19
- Said S. E., Dickey D. A., 1984, *Biometrika*, 71, 599
- Sanidas S. et al., 2019, *A&A*, 626, A104
- Scheuer P. A. G., 1968, *Nature*, 218, 920
- Slee O. B., Dulk G. A., Otrupcek R. E., 1980, *PASA*, 4, 100
- Stappers B. et al., 2011, in Burgay M., D'Amico N., Esposito P., Pellizzoni A., Possenti A.eds, AIP Conf. Ser. Vol. 1357, Radio Pulsars: An Astrophysical Key to Unlock the Secrets of the Universe. American Institute of Physics, p. 325
- Stinebring D. R., McLaughlin M. A., Cordes J. M., Becker K. M., Goodman J. E. E., Kramer M. A., Sheckard J. L., Smith C. T., 2001, *ApJ*, 549, L97
- Stinebring D. R. et al., 2022, *ApJ*, 941, 34
- Walker M. A., Melrose D. B., Stinebring D. R., Zhang C. M., 2004, *MNRAS*, 354, 43
- Williamson I. P., 1972, *MNRAS*, 157, 55
- Williamson I. P., 1973, *MNRAS*, 163, 345
- Wu Z. et al., 2022, *A&A*, 663, A116
- Wu Z. et al., 2023, *MNRAS*, 520, 5536
- van Haarlem M. P. et al., 2013, *A&A*, 556, A2
- van Straten W., Bailes M., 2011, *PASA*, 28, 1
- van Straten W., Demorest P., Osłowski S., 2012, *Astron. Res. Technol.*, 9, 237

APPENDIX A: SCATTERED PULSE PROFILES

In this appendix we present the scattered pulse profiles for the remaining five pulsars from our project (see Fig. 1). The profiles are taken from individual observations with 3-h integration times and split into six sub-bands to obtain separate scatter time measurements. In some of the cases shown, the highest frequency band was omitted, due to poor profile quality caused by the combination of the effects of pulsar spectra (pulsars are weaker at higher frequencies), and heavy interference.

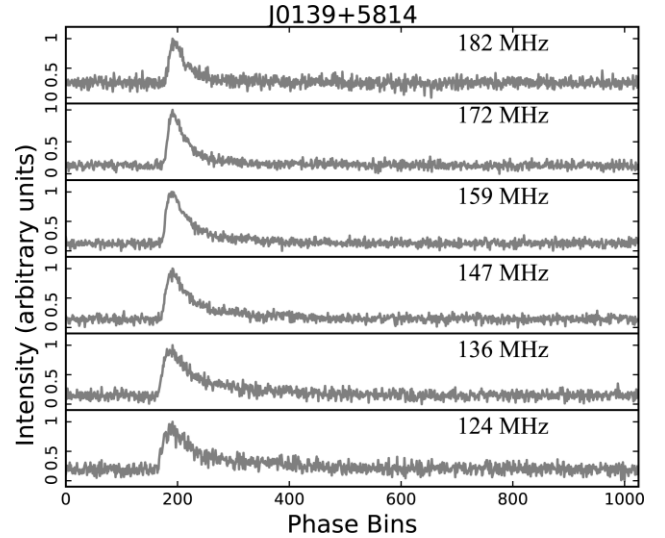


Figure A1. An example set of profiles of PSR J0139+5814 from a 3-h observation from the PL-611 LOFAR station. The observing bandwidth of 72 MHz was split into six sub-bands.

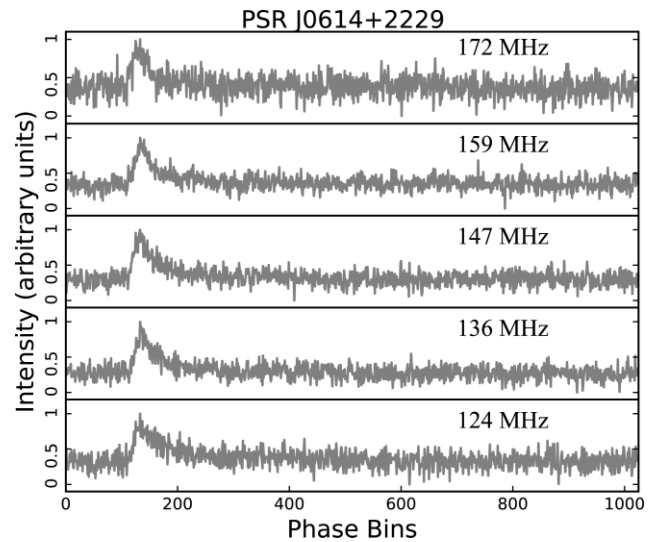


Figure A2. An example set of profiles of PSR J0614+2224 from a 3-h observation from the PL-611 LOFAR station. The observing bandwidth of 72 MHz was split into six sub-bands. One of the resulting sub-bands was omitted due to poor quality profile and heavy interference.

APPENDIX B: INTRINSIC PULSE WIDTHS

As it was mentioned in Section 1 the observed pulse profile is a convolution of the intrinsic pulse profile and the scattering model dependent PBF. During our modelling of the scattered profiles we obtained the values of the scatter time τ from each frequency sub-band of each observation, but also the associated value of the width of the intrinsic pulse profile (see Section 2.2). The intrinsic pulse width was treated as one of the parameters of the fit, to allow for its adjustment at different observing frequencies, which is necessary due to the effect of RFM (Kijak & Gil 2003), i.e. the natural broadening of the pulse profiles at lower frequencies due to different emission height at different frequencies. In this appendix we show the plots of the intrinsic pulse widths versus the observing frequency for selected observations (one per pulsar) that we obtained during our modelling

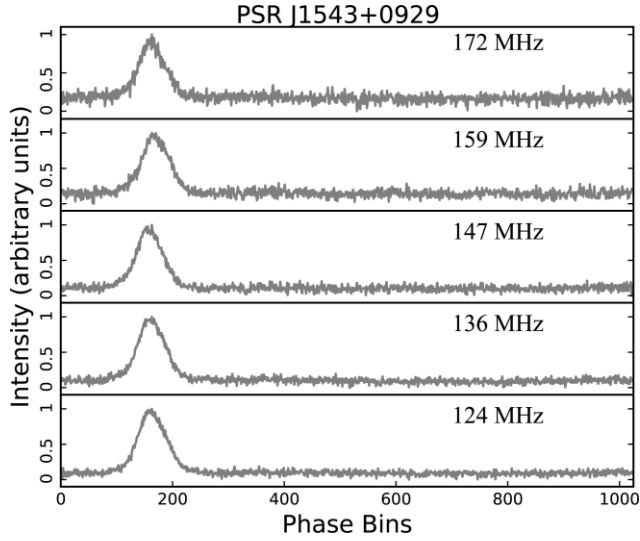


Figure A3. An example set of profiles of PSR J1543+0929 from a 3-h observation from the PL-611 LOFAR station. The observing bandwidth of 72 MHz was split into six sub-bands, however the highest frequency sub-band had to be removed due to poor quality, as the profile's S/N was heavily affected by the removal of a number of spectral channels due to radio interference.

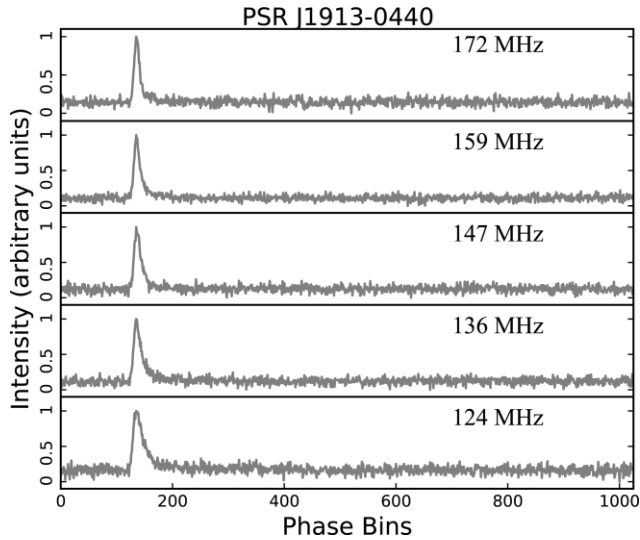


Figure A4. An example set of profiles of PSR J1913-0440 from a 3-h observation from the PL-611 LOFAR station. The observing bandwidth of 72 MHz was split into six sub-bands. One of the resulting sub-bands was omitted due to poor quality profile caused by RFI and the removal of a number of frequency channels.

of the scattered profiles. These plots show the values of the widths (usually denoted as w_{50} i.e. full width at half-maximum) of the fitted intrinsic Gaussian profile. A clear intrinsic pulse width evolution is noticeable for PSR J0139+5814 and PSR J2113+4644, and possibly less obvious for pulsars J1913-0440 and J1917+1353, where the uncertainties of individual estimates are relatively large. No expected RFM effects are visible for PSR J0614+2224 and PSR J1543+0929, which indicates that during the modelling some (or all) of the intrinsic pulse width changes were misinterpreted as scattering – which we

mention in the individual pulsar discussions (see Section 3.1) for these two objects.

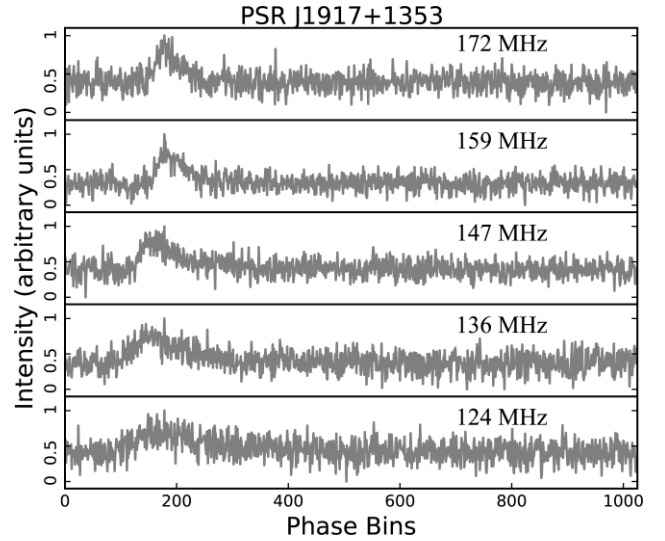


Figure A5. An example set of profiles of PSR J1917+1353 from a 3-h observation from the PL-611 LOFAR station. The observing bandwidth of 72 MHz was split into six sub-bands. One of the resulting sub-bands was omitted due to poor quality profile caused mainly by heavy radio interference.

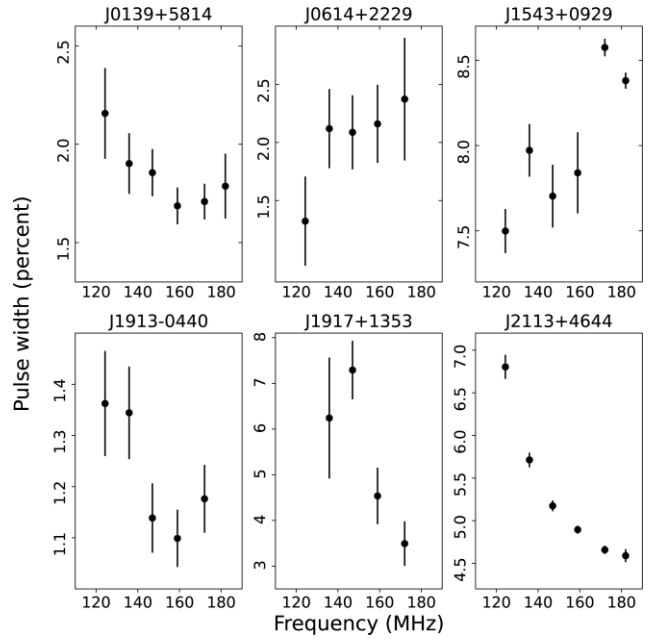


Figure B1. Plots showing the values of the intrinsic pulse widths (shown as the percentage of the pulse period) versus the observing frequency for a selected observation for each of the six pulsars from our project. We obtained these values as the FWHM of the Gaussian approximation of the intrinsic profile that was used during our modelling of the scattered pulse profiles (Section 2.2). See the Appendix text for more detailed explanation.

This paper has been typeset from a $\text{\LaTeX}$ file prepared by the author.